

1.1.7 Advanced Logic Gates (XOR and XNOR)

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔

CS XI

Chapter 1: Computer Systems

Lecture 09 of 20

1.1.7 Advanced Logic Gates (XOR and XNOR)

LEARN IT

XOR (Exclusive OR) gives output 1 when the inputs are **different**. Expression: $Y = A \oplus B$. In terms of basic operations, $A \oplus B = A'B + AB'$.

XNOR (Exclusive NOR) is XOR followed by NOT. It gives output 1 when the inputs are the **same**. Expression: $Y = (A \oplus B)'$ = $A'B' + AB$.

The XOR symbol is an OR symbol with an extra curved line at the input side. XNOR adds a bubble at the output.



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

$$\text{XOR — } Y = A \oplus B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

$$\text{XNOR — } Y = (A \oplus B)'$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

All seven gates at a glance

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

(NOT has one input: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.)

WORKED EXAMPLE

A staircase light. A light has one switch at the bottom (A) and one at the top (B) of a staircase. Flipping either switch should change the light.

Step 1: Start with both switches down: $A = 0$, $B = 0$, light off ($Y = 0$).

Step 2: Flip the bottom switch: $A = 1$, $B = 0 \rightarrow$ light on ($Y = 1$).

Step 3: Walk up and flip the top switch: $A = 1$, $B = 1 \rightarrow$ light off ($Y = 0$).

Step 4: The pattern 0, 1, 1, 0 is the XOR truth table. So the light is $Y = A \oplus B$.

REMEMBER IT

Memory trick: XOR outputs 1 when inputs are "**eX**clusively" different. XNOR outputs 1 when inputs are the **same**. XNOR is sometimes called the "equality gate".

COMMON MISTAKE

XOR is not the same as OR. They differ only in the last row: for $A = 1, B = 1$, OR gives 1 but XOR gives 0.

اردو میں سمجھیں

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔
XNOR دراصل XOR کے بعد NOT ہے۔ OR اور XOR میں فرق صرف آخری قطار میں ہے جہاں دونوں ان پٹ 1 ہوں۔

CHECK YOURSELF

1. What is $1 \oplus 1$?
2. Which gate outputs 1 when both inputs are the same?
3. Write XOR using AND, OR and NOT.

Answers: 1. 0. 2. XNOR. 3. $A \oplus B = A'B + AB'$.

Next lecture: 1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms